3 A sample of gas contains a fixed amount of gas. The gas has pressure p, volume V and thermodynamic temperature T.

Fig. 3.1 shows the variation of pV with kT for the sample, where k is the Boltzmann constant.

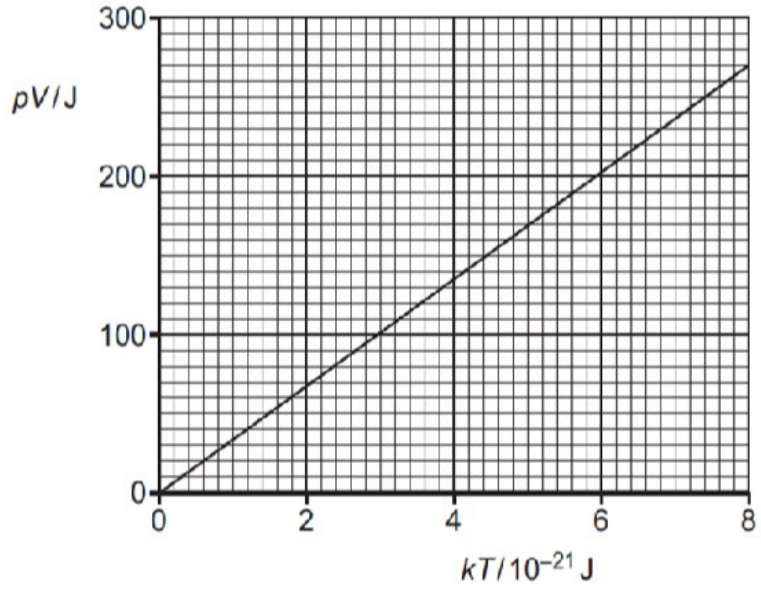


Fig. 3.1

(a) **(i)** Explain what is indicated about the nature of the gas from the variation shown in Fig. 3.1.

.....

.....

..... [1]

(ii) Determine the amount n of molecules of the gas in the sample.

n = ... mol [2]

(b) The root-mean-square (r.m.s.) speed of the molecules of the gas is 1900 m s^{-1} when pV is equal to 270 J.

Determine the mass of one molecule of the gas.

mass = ... u [3]